

**DEPARTMENT OF COMPUTER
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COURSE TITLE: COMPUTER
APPLICATION ONE
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Course Outline

1. Know the existing application packages.
2. Understand word processing packages.
3. Know electronic spread sheets.
4. Know the fundamentals of accounting packages.
5. Understand presentation packages.
6. Know how to use education, medical and other packages.

What is Software?

Software is a set of instructions, data, or programs used to operate computers and execute specific tasks. It's the intangible part of a computer system, contrasting with hardware, which is the physical components.

Examples: This includes operating systems (like Windows or mac OS), applications (like web browsers, word processors, or games), and programming tools.

Or

Software is a set of instructions that tells a computer, web-based application, or other devices what to do. It helps the device understand what to do and how to do it. Through software, you can interact with the device in ways they couldn't before. For example, with software, you can use a computer to create graphics, edit videos, create music, and play games without having to know how the hardware works. The software makes interacting with computers easier and allows us access to new features and capabilities not possible without it.

Types of Software

- i. System software
- ii. Application software

Difference between System Software, Program Generators and Application Packages

System Software	Program Generator	Application Package
i. System Software is a type of computer program that manages and controls a computer's hardware and application software. ii. It acts as an interface between the hardware and the application software, allowing users to interact with the computer. Examples include operating systems (OS), device drivers, and utility software.	i. Program Generator also known as a code generator, is a tool that automates the creation of computer programs or code segments. ii. It takes specific inputs, like templates or rules, and generates corresponding code in a target programming language.	i. Application packaging refers to the process of preparing software applications for distribution and deployment across various computing environments. ii. It involves bundling all the necessary files, configurations and dependencies into a standardized package that can be easily installed and managed.

Identify the Modes of Package Acquisition

Software packages can be acquired through the following ways:

- i. Buying from Software Developers
- ii. Renting from Software Houses
- iii. Buying from Software Vendors
- iv. Hiring from Software Developers

State the Criteria for Package acceptability: The following are Criteria for Software Acceptability.

1. Whether the Packages meet the users needs
2. Whether the packages meet up with the processing requirements
3. Whether the packages are affordable by those who need them

4. Whether the packages are reliable

Define Word Processor

A word processor is a device or software program capable of creating, storing, and printing text documents. It allows users to write and modify text, display it on a screen, save it electronically, and print it out.

Word processing software is one of the most common technology tools in the world, allowing users to create resumes and cover letters, business correspondence, blog posts, novels, and more. Traditionally, word processors were computer applications downloaded to a specific number of computers. But as cloud computing has become more common, so have browser-based word processors. These typically lack the more advanced features of a traditional program; however, they offer more flexibility and the ability to collaborate in real time.

State the Advantages of Word Processor

i. Easy Editing

Word processors allow for effortless modifications, insertions, deletions, and rearrangements of text.

ii. Spell and Grammar Checking

Built-in tools help identify and correct errors, ensuring professional-quality documents.

iii. Formatting Options

Users can customize fonts, styles, spacing, and layouts to create visually appealing and readable documents.

iv. Saving and Sharing

Documents can be saved electronically and shared easily via email or cloud storage, facilitating collaboration.

v. Efficiency

Word processors allow for faster typing and editing compared to traditional methods like handwriting or typewriters.

vi. Content Management

Features like inserting images, tables, and charts enhance document content and readability.

vii. Collaboration:

Features of the Main, Help and Others Menus

Word processors enable multiple users to work on the same document simultaneously, facilitating real-time collaboration. Word processors like Microsoft Word or Google Docs, while primarily used for document creation, often integrate with other professional software packages like desktop publishing tools (CorelDraw, PageMaker) for specific functions. These functions include text input, editing, formatting, and importing text into the DTP environment, providing a foundation for creating professional-looking layouts.

Identify Functions of Word Processors in other Professional Packages

1. Creating content

The core function of word processors is to allow users to type and edit text, which forms the basis of many desktop publishing projects.

2. Saving and organizing:

Word processors offer features to save and organize documents, making it easy to manage the content before importing it into a DTP application.

3. Basic text formatting:

Word processors allow users to apply basic text formatting like font type, size, color, bold, italics, and underlining, which can be applied to text imported into DTP software.

4. Page layout:

While not as extensive as DTP software, word processors also offer some basic page layout options like margins and headings.

Importing text:

Word processors often export documents in formats compatible with DTP software, such as .doc, .docx, or .rtf, allowing users to import the text into their layout programs.

5. Text manipulation:

Once imported, the DTP software can further refine the text, adjusting fonts, spacing, and paragraph styles to achieve the desired visual effect.

Explain use of document and Non document.

- ✓ The main document contains the basic text that is the same in all of the output documents. It may contain a letterhead, text, and instructions in merge fields for inserting text (such as recipient names and addresses) that vary from one output document to another.
- ✓ Non-document text processing refers to tasks that involve manipulating and analyzing text that isn't typically found within a standard document format. This includes tasks like data extraction from various sources, text summarization, and more. Mail merging is a specific type of non-document processing that combines a document template with a separate data source to create personalized outputs, like letters or emails.

What is Import Graphics and the Creation of drawing Objects?

Importing graphics refers to bringing visual elements like images, illustrations, or designs into a digital document or application, such as a word processor, photo editor, or design software. This process allows users to incorporate external visual content, like a photo, into a project, creating a visual impact or adding information to their work.

Drawing objects refer to the various graphical elements that you can create and manipulate in a document, presentation, or graphic design software. These objects can include shapes, lines, text boxes, images, charts, and other visual elements that enhance the content and layout of your document or presentation. Drawing objects allow you to add visual appeal, structure, and organization to your work.

In many software applications, including word processors like Microsoft Word and presentation software like Microsoft PowerPoint, you can find a Drawing toolbar or a set of drawing tools that help you create and customize drawing objects. The Drawing toolbar typically provides various

options and tools for working with drawing objects. Here are some common options and tools you might find on a Drawing toolbar:

Shape Tools: These tools allow you to create different shapes such as rectangles, circles, arrows, lines, and more. You can select a shape tool, draw the shape on your document or slide, and then customize its size, color, and style.

Text Box: You can add text to your document or presentation by using a text box tool. This tool lets you create a box where you can enter and format text. Text boxes are useful for adding labels, titles, and captions to your drawings.

Line and Arrow Tools: These tools help you draw straight lines or arrows with different styles and thicknesses. Lines and arrows are useful for creating flowcharts, diagrams, or highlighting specific content.

Fill and Line Color: You can choose the fill color (the interior color of a shape) and line color (the color of the shape's border) to customize the appearance of your drawing objects.

Text Formatting: When you're working with text inside a drawing object, you can use text formatting options to change the font, size, alignment, and style of the text.

Grouping and Arrangement: These options allow you to group multiple drawing objects together, align them, distribute them evenly, and control their layering or stacking order.

Rotation and Flip: You can rotate drawing objects to different angles or flip them horizontally or vertically to achieve the desired orientation.

Locking and Protection: Some software provides options to lock or protect drawing objects to prevent accidental modifications.

Insert Picture or Image: You can insert images or pictures from your computer or external sources into your document or presentation.

Explain Sharing Data with Other Users

Sharing data with other users involves making data accessible to people beyond the individual who owns or controls it. This could be within an organization, with external partners, or even with the public. The specific process and reasons for sharing can vary greatly.

Method of sharing data

1. Via Social Media
2. Bluetooth
3. Xender
4. Flash drive
5. Memory Card
6. External Hard drive
7. Via Mail

Types of existing spread sheets.

Desktop Spreadsheets

These are installed on a user's computer and are often part of a larger software suite (like Microsoft Office).

Web-based Spreadsheets

These are accessed through a web browser and are often part of cloud-based productivity suites (like Google Workspace).

Spreadsheet-Database Hybrids

These combine the features of a spreadsheet and a relational database, allowing for more structured data management.

Project Management Spreadsheets

These focus on project tracking, task management, and collaboration, often with additional features like Gantt charts.

What is Spreadsheet?

A spreadsheet is a computer program that can capture, display and manipulate data arranged in rows and columns. Spreadsheets are among the most popular tools available for computer users.

Or

A spreadsheet is generally designed to hold numerical data and short text strings. In a spreadsheet program, spaces that hold items of data are called spreadsheet cells. Cells can be renamed to reflect the data they hold and can be cross-referenced through row numbers and column letters.

Explain the use of spread sheet in a forecasting project

The true power of the spreadsheet is its ability to handle complex mathematical calculations and automatically recalculate totals as the underlying data in the sheet changes. This is especially useful for "What-if" analysis and forecasting.

Explain the use of spread sheet in a forecasting financial analysis

Spreadsheets are crucial in financial forecasting and analysis because they allow users to organize, analyze, and present financial data in a structured way. This enables businesses to make informed decisions by visualizing data, identifying trends, and simulating various scenarios.

What is the use of spread sheet in a forecasting production scheduling?

Spreadsheets are valuable tools in production scheduling and forecasting. They help organize and analyze data, allowing for efficient tracking of inventory, scheduling of production runs, and forecasting demand. Spreadsheets can also be used for "what-if" analysis, allowing users to see the impact of different scenarios on production and inventory.

Explain how to carrying out general statistical functions using cell references in a spreadsheet.

To perform statistical functions in a spreadsheet like Microsoft Excel using cell references, you first select the cell where you want the result to appear. Then, type an equal sign (=), followed by the desired function name (e.g., AVERAGE, SUM, COUNT). Next, enclose the cell references or range of cells within parentheses. You can specify a range of cells using the colon (:) or separate cells with commas. Finally, press Enter to calculate and display the result in the selected cell.

Explain performing specific accounting functions using spread sheets and highlight data security requirements on spread sheet data.

Spreadsheets, like Excel, are commonly used in accounting for tasks like budgeting, financial analysis, and data recording. They offer features like formulas, sorting, and filtering, enabling

efficient analysis of financial data. Data security is crucial; measures like worksheet protection, password protection, and limiting access can safeguard sensitive accounting information.

Specific Accounting Functions in Spreadsheets:

1. **Budgeting:** Spreadsheets facilitate creating budgets by tracking income and expenses, allowing users to analyze spending patterns and make projections, says Babington.
2. **Financial Analysis:** Analyzing financial data, like profit margins, by using formulas, pivot tables, and charts, helps understand financial performance and identify areas for improvement, notes Go Fish Digital.
3. **Data Recording and Organization:** Spreadsheets can be used to record transactions, invoices, and other financial records, offering a centralized and organized way to store and manage financial data, explains U.S. Environmental Protection Agency (.gov).
4. **Calculations and Reporting:** Spreadsheets enable quick calculations and generate reports, such as income statements and balance sheets, providing a visual overview of financial health.

Data Security Requirements

Worksheet Protection: Limiting access to specific areas of the spreadsheet by locking certain cells, prevents unauthorized modifications, according to Microsoft Support.

Password Protection: Protecting the entire spreadsheet with a password prevents unauthorized access and ensures that only authorized personnel can view or modify the data.

Access Control: Implementing user access restrictions to limit who can view, edit, or share the spreadsheet, further enhances data security.

Data Backups: Regularly backing up spreadsheets to prevent data loss in case of hardware failure or accidental deletion.

Data Encryption: Consider encrypting sensitive data within the spreadsheet, especially when sharing or storing the file externally.

Explain formatting worksheets and working with formulas

Formatting worksheets involves changing the appearance of a worksheet to enhance readability and highlight important data, while working with formulas enables calculations and manipulation

of data within the worksheet. Formatting can include font styles, colors, alignment, and borders, while formulas use cell references and mathematical operators to perform calculations.

Formatting Worksheets

Font: Change font style, size, color, and effects like bold, italic, and underline.

Alignment: Adjust text alignment within cells (left, right, center, top, bottom).

Number Format: Apply formats like currency, percentage, date, or create custom formats.

Borders: Add or modify borders around cells or ranges.

Fill Color: Change the background color of cells or ranges.

Styles: Use predefined styles or create custom styles for consistent formatting.

Conditional Formatting: Apply formatting based on cell values or formulas, highlighting data that meets certain criteria.

Themes: Apply document themes for a uniform look throughout the worksheet.

Explain performing specific accounting functions using spread sheets

Spreadsheets, like Microsoft Excel, are versatile tools that can be used for various accounting functions. They allow for tracking income and expenses, managing product information, creating budgets and forecasts, and analyzing cash flows. Spreadsheets also provide tools for creating financial statements, such as balance sheets and income statements.

Highlight data security requirements on spread sheet data

To ensure data security in spreadsheets, it's crucial to implement measures like encryption, limiting access, using data validation, backing up data, and auditing activity. Specifically, password protect the file, apply sensitivity labels, protect cells to prevent accidental changes, hide sensitive information, and utilize data validation to ensure data integrity.

Ways to protect your spreadsheet

1. Encryption:

- **Protect the Workbook:** Use the "Protect Workbook" feature in Excel to encrypt the file with a password, preventing unauthorized access.

- **Password Strength:** Create strong passwords that combine uppercase and lowercase letters, numbers, and special characters.

- **Encryption Algorithms:** Ensure that the encryption method used is robust and secure.

2. Access Control and Sharing:

- **Limit Access:** Control who can open, edit, and share the spreadsheet.
- **Read-Only Access:** Grant read-only access to users who only need to view the data.
- **Restrict Sharing:** Use the "Share" feature in Excel to restrict access to specific individuals or groups.

3. Data Validation and Protection:

- **Data Validation:**

Use data validation rules to restrict input to specific formats and ranges, preventing errors and inconsistencies.

- **Protect Cells:**

Lock specific cells or areas of the worksheet to prevent unauthorized changes.

- **Unlock Specific Areas:**

Allow users to edit specific cells or areas of the worksheet while preventing changes to other areas.

4. Backup and Updates:

- **Regular Backups:** Create regular backups of your spreadsheets to protect against data loss.
- **Version Control:** Utilize version control features to track changes and revert to previous versions if needed.
- **Security Updates:** Keep your spreadsheet software (e.g., Excel) updated with the latest security patches.

5. Audit and Monitoring:

- **Audit Trail:** Enable auditing to track changes and access to the spreadsheet.
- **Monitoring Activity:** Monitor user activity to detect potential security breaches or suspicious behavior.

- **Regular Reviews:** Periodically review security settings and access permissions to ensure they remain appropriate.

Formatting worksheets enhances readability and data presentation, while formulas perform calculations. Formatting options include changing font, alignment, borders, and colors, while formulas use cell references and mathematical operators to calculate values.

Formatting Worksheets

Formatting worksheets allows you to customize the appearance of your data for clarity and visual appeal. This includes:

- **Font, Alignment, and Borders:**

You can change the font type, size, color, and alignment of text within cells, add borders to create visual separation, and even use shading to highlight specific data.

- **Number Formats:**

Apply number formats like currency, percentage, date, or custom formats to ensure data is displayed consistently and accurately.

Formatting Worksheets and Working with Formulas

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- **Column Widths and Row Heights:**

Adjust the width of columns and height of rows to accommodate different text lengths and data types.

Transfer of Information and Graphics between Applications

Transfer of Information refers to the process of moving digital information between systems, devices, or locations, or applications enabling communication and access to data across networks and server_storage_solutions. Data transfer can occur within a single computer system, between local devices (for example, external hard drives or USBs), or across vast global networks, such as the internet. It is fundamental to the functionality of computer networks, cloud computing, data storage solutions, and virtually every aspect of digital communication.

Accounting Packages

Desktop-based Software:

These are traditional software packages that are installed on a user's computer. They can be a good option for businesses that need more control over their data and prefer to work offline.

Cloud-based Software:

These are web-based applications that can be accessed from any device with an internet connection. They offer flexibility, scalability, and collaboration features, making them a popular choice for businesses of all sizes.

On-premises Software:

These are software packages that are installed on the company's own servers. They offer a high level of control and security, but also require more technical expertise to manage.

Software as a Service (SaaS):

These are cloud-based software solutions that are accessed through a subscription. They offer a convenient and cost-effective way to access accounting software.

ERP Software:

These are comprehensive enterprise resource planning systems that integrate accounting with other business functions, such as supply chain management and human resources.

Define Accounting

Accounting is a process of recording, classifying, summarizing, and reporting financial transactions of a business or organization. It involves tracking, organizing, and interpreting a company's financial activities to manage cash flow, prepare reports, and support decision-making. Accounting provides a systematic approach to understanding a company's financial health and performance.

Define Financial Management

Financial management encompasses the planning, organizing, directing, and controlling of a company's financial resources to achieve its goals. It involves making strategic decisions related to funding, investments, and cash flow, all while aiming to maximize profits, minimize risks, and ensure compliance with regulations. Essentially, it's about managing money effectively to ensure the long-term financial health and success of an organization.

Areas in Accounting where Accounting Packages is been used

1. General Ledger (GL)

- Core component of most accounting systems.
- Records all financial transactions.
- Enables preparation of financial statements (income statement, balance sheet, etc.).

2. Accounts Payable (AP)

- Manages what the company owes to suppliers.
- Tracks vendor invoices, due dates, and payments.
- Automates bill entry and payment processing.

3. Accounts Receivable (AR)

- Tracks customer invoices and incoming payments.
- Automates invoicing, receipt tracking, and follow-ups for overdue accounts.

4. Bank Reconciliation

- Matches company's bank statements with internal records.
- Helps detect errors or fraudulent activity.
- Often includes import tools for bank feeds.

5. Payroll Accounting

- Calculates employee wages, deductions, and taxes.
- Manages payroll tax reporting and direct deposits.
- Ensures compliance with labor and tax laws.

6. Inventory Management

- Tracks inventory levels, purchases, and cost of goods sold (COGS).
- Integrates with sales and purchasing systems.

7. Fixed Asset Management

- Manages company's fixed assets like machinery, vehicles, and buildings.
- Tracks depreciation, asset lifecycle, and disposals.

8. Budgeting and Forecasting

- Allows creation of budgets and comparison with actual results.
- Supports scenario analysis and long-term planning.

9. Financial Reporting

- Generates standard and custom reports.
- Includes profit and loss, balance sheet, cash flow statements.
- Often includes tools for analysis and visualization.

10. Tax Management

- Calculates and tracks tax liabilities (VAT, sales tax, etc.).
- Helps generate tax returns and reports.

- Often includes compliance features based on local laws.

11. Project/Job Costing

- Tracks income and expenses by project or job.
- Useful for service-based businesses (consulting, construction).

12. Multi-currency and Multi-entity Management

- Supports companies operating in multiple countries.
- Handles currency conversion and consolidation of accounts.

Types of Existing Accounting Packages

1. Cloud-Based Accounting Software
2. Desktop Accounting Software
3. Enterprise Resource Planning (ERP) Systems
4. Payroll Software

Facilities that make each packages

Material: Emphasize the quality, sustainability, or unique properties of the packaging material. For example, if using recycled materials, highlight their eco-friendly nature. If using a premium material, emphasize its luxurious feel or durability.

Design: Showcase the packaging's visual appeal, including colors, fonts, graphics, and overall aesthetic. Highlight how the design complements the brand identity and attracts the target audience.

Functionality: Describe how the packaging makes the product easier to use, store, or protect. For example, highlight features like resealable closures, tamper-evident seals, or child-resistant packaging.

Brand Messaging: Integrate the brand's values and story into the packaging design. Highlight the brand's commitment to quality, sustainability, or customer satisfaction. Use the packaging to build trust and establish a positive brand image.

Unique Features: Emphasize any unique or innovative aspects of the packaging. This could include interactive elements, personalized designs, or augmented reality features.

Specific Examples: Retail: Highlight the visual appeal of the packaging and how it draws attention on the shelf.

E-commerce: Emphasize the durability and protection provided by the packaging during shipping.

Food and Beverage: Highlight the freshness and safety guarantees of the packaging.

What is payroll, job costing, and invoicing and order process?

Payroll refers to a software system or program designed to automate and manage the process of paying employees. It handles everything related to employee compensation, including calculating wages, taxes, and deductions, and issuing payments through methods like direct deposit or checks.

Job costing is a method of tracking and allocating costs to specific projects or jobs. It's essentially a way to determine the total cost of producing a unique product or service, rather than an average cost across a large volume of goods.

Invoicing refers to the process of creating, sending, and managing invoices electronically using software or online systems. This involves generating invoices, tracking their status, and facilitating online payments, all while minimizing manual tasks.

Order processing refers to the systematic workflow of accepting, managing, and fulfilling customer orders. It's a series of steps that begins when a customer places an order and continues through inventory checks, picking and packing, shipping, and tracking until the product is delivered.

Functions of a Presentation Package using Power Point

- Creation of slides: slides titles are used to define the information on the slide as well as create a navigational outline for the presentation.
- Insertion of pictures: Pictures can be inserted into the presentation by using a layout into the presentation by using a layout with a picture place holder shape, pictures can also be used as background for presentation graphics.
- Insertion of video and audio: some presentation packages include objects such as sound clip animated graphics, video clips or other media objects, these objects are generally added to

emphasize the content of the presentation or provide an alternative means of delivering information.

- Animation: Customize animation is often added to a slide show. It can be used to delay the appearance of bullet points on the screen. The can help to keep audience focused on what is being presented.
- Slide show: the slide show has different views, they are: From beginning or F5, from current slide or shift + F5
- Custom slide: This creates or plays a custom slide show displays only the slides you select.
- Set up slide show: It shows set up advance option such as kiosk mode
- Hide slide: this will hide current slide from the presentation from showing during the full.
- Slide transition: this can be set to govern how one moves from one slide to another
- Creation of graphics: graphics images can be created using elements of type, imagery, colour and shape in order to communicate a message or idea visually.

Types of Presentation

1. Educational Presentations

Educational presentations are a great way to introduce a new topic to people who aren't familiar with it. They can be especially helpful when you want to explain something complex, like a process, or share important facts in a clear way. Whether you're a teacher in a classroom or a trainer at a company, educational presentations can be a powerful tool for learning.

2. Instructional Presentations

Instructional presentations help people learn more about a topic and sometimes even guide them on what to do next. They're similar to presentations used for education, but they might include some extra details or specific steps for the audience to follow.

3. Persuasive Presentations

Many presentations aim to convince the audience of something, like a new idea, product, or way of doing things. They often address a specific issue and use facts and figures to explain why their solution is the best. Business proposals and sales talks are common examples.

4. Motivational Presentation

Motivational presentations aim to lift up the audience's spirits and help them deal with challenges. They spark interest in a subject and share a particular perspective or message. These presentations can be useful when you want to inspire a group of people.

5. Problem-solution Presentation

Have you ever needed to present an idea to help people make a decision? Problem-solution presentations are a great way to do that. They focus on explaining a challenge or issue and then offering potential solutions for the audience to consider. While similar to persuasive presentations, the main goal here is to discuss the problem clearly and share research so decision-makers can weigh the options and choose the best path forward. These presentations can include details about the problem and a few possible solutions. They're a handy tool for many business meetings and discussions within organizations.

6. Project Presentations

Progress presentations are a way to share how a project, campaign, or initiative is moving along. They're similar to progress reports, but in a presentation format.

7. Storytelling Presentations

Presentations that use a story format can be a great way to connect with your audience and share information in a more engaging way. This approach can be useful in many settings, from classrooms to company meetings. It can be especially helpful when you want to grab the attention of a specific group of people and make them feel involved.

8. Visual Presentations

Presentations come in many forms, but some rely mostly on pictures, charts, and other visuals instead of text. These are called visual presentations. They're a good choice when you have limited time or your topic is easy to understand with pictures.